Type MS · Plan an investigation measuring a response or treatment in people

Answer 1 of 14 · Question 5 part b.ii · 6 marks

2023 JUN · Paper 02 · extract 1 of 2 · find part b.ii in this mark scheme extract

Number				
5	(a)	*hepatic vein* vena cava; pulmonary artery; pulmonary vein; aorta; renal artery;	Credit correct order	5
	(b) (i)	Any four from: • as time spent exercising increases, pulse rate increases; • increase flow of blood (to muscles/cells); • more oxygen/glucose/energy; • more (aerobic) respiration; • more carbon dioxide removed/reference to reduced build-up of lactic acid;	Allow more blood pumped Allow less anaerobic respiration	4
	(ii)	Any six from • take pulse before exercise/resting pulse; • finger at wrist/neck; • carry out exercise; • count/record pulse (after exercise); • wait for pulse rate to drop back to normal; • repeat exercise; • reference to control variables e.g. same person/gender/exercise/intensity of exercise/weather conditions;	Allow reference to use of electronic device to measure pulse rate Allow adequate time frame e.g. 5 minutes Allow reference to reliability e.g. more people	6

Total 15

Type MS · Plan an investigation measuring a response or treatment in people

Answer 1 of 14 · Question 5 part b.ii · 6 marks

2023 JUN · Paper 02 · extract 2 of 2 · find part b.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 2 of 14 · Question 2 part b · 4 marks

2020 JAN · Paper 02 · find part b in this mark scheme extract

Number			
2 (a) (i)	column headings(1) units in column headings(1) values correctly written in columns(1) Type of activity / exercise: at rest; walking; jogging; running Pulse/heart rate in beats per minute: 78; 95; 138; 146	Allow bpm for beats per minute	3
(b)	Any four from: • take pulse rate at rest(1) • using monitor/from wrist/neck(1) • walk (for one minute)(1) • take pulse rate after walking(1) • repeat for jogging and running(1) • rest between each exercise(1)		4
(c)	exercise for the same amount of time/same terrain/environmental conditions/wear same clothing		1
(d)	breathing/metabolism/respiration/blood pressure/sweating		1

Total for Question 2 = 9 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 3 of 14 · Question 4 part a.iii · 4 marks

2023 JUN · Paper 02 (regional R) · find part a.iii in this mark scheme extract

<table>
<tr>
<td>4 (a) (i)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td></td></tr>
<tr><td>jogging</td><td></td></tr>
<tr><td>running</td><td></td></tr>
</table>
</td>
<td>1 mark for valid table
1 mark for labels;
1 mark for units;</td>
<td>3</td>
</tr>
<tr>
<td>(ii)</td>
<td>
<table>
<tr><th>Activity</th><th>Pulse rate
in beats per minute</th></tr>
<tr><td>walking</td><td>94</td></tr>
<tr><td>jogging</td><td>116</td></tr>
<tr><td>running</td><td>133.5</td></tr>
</table>
</td>
<td>2 marks for all readings correct.
1 mark for 1 or 2 readings correct.

tolerance +/- ½ square</td>
<td>2</td>
</tr>
<tr>
<td>(iii)</td>
<td>Any four from five

• take resting pulse;
• method for measuring pulse rate e.g. finger on wrist/neck/electronic device;
• carry out an activity/exercise (for a set period of time/given time);
• measure pulse rate after activity;
• repeat (for each activity);</td>
<td></td>
<td>4</td>
</tr>
<tr>
<td>(b)</td>
<td>Any four from

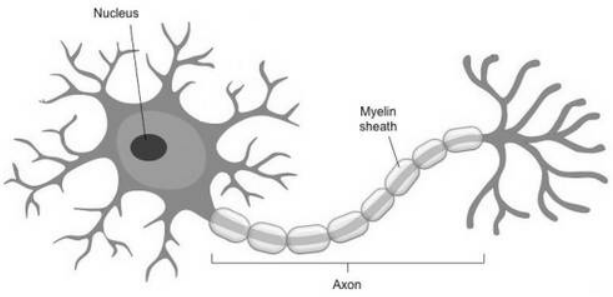
• faster/more/greater muscle contraction during running;
• more energy/ATP required;
• faster rate of respiration;
• more oxygen/ glucose required/more CO_2 removed;
• increase blood flow/faster blood flow/more blood (to muscles);</td>
<td>ORA for all marking points</td>
<td>4</td>
</tr>
</table>

Total for Question 4 = 13 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 4 of 14 · Question 3 part b.i · 6 marks

2025 JUN · Paper 02 (regional R) · find part b.i in this mark scheme extract

3	(a)	(i)		1 mark for diagram 3 marks for labels	4
		(ii)	• transmits electrical/nerve impulses; • from CNS/brain/spinal cord/relay neurone, to effector organs/named effector organ;		2
	(b)	(i)	Any six from 1 (student X) holds up the ruler between the thumb and forefinger (of other student); 2 then drops/releases the ruler; 3 (other) student catches the ruler; 4 record the reading on the ruler/distance fallen when caught; 5 repeat for <u>this</u> student; 6 using the same distance between thumb and forefinger; 7 find mean/average value of reading on ruler; 8 calculate reaction time (using formula); 9 repeat with <u>other</u> students;		6
		(ii)	the reading on the ruler/ reaction time;		1
		(iii)	independent variable not continuous/is discrete;		1
		(iv)	reaction time = √(0.15 ÷ 5); reaction time = 0.17 (s);	accept 0.173(2)	2
				Total for Question 3 = 16 marks	

Type MS · Plan an investigation measuring a response or treatment in people

Answer 5 of 14 · Question 3 part c.ii · 2 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part c.ii in this mark scheme extract

Number				
3	(a)	any two from • protection/named example/barrier against pathogens; • sensory reception/named example; • excretion		2
	(b)	• looks red; • vasodilation/more blood in skin <u>capillaries</u>/blood flows closer to skins surface; • allows more heat loss/heat energy released; • to lower body temperature;		4
	(c) (i)	• as temperature increases, rate of sweating increases; • up until 34°C/ between 32 °C - 35.9 °C, little/no change; • increases, rapidly/significantly from 35.9 °C - 38 °C;	Allow any temperature between 34°C and 35.9 °C for 35.9 °C e.g. between 32°C and 34 °C	2
	(ii)	• absorb sweat with cotton wool/absorbent paper; • fixed area of skin/under arm pits; • reference to measuring mass of absorbent material before and after; or • measure time; • for cobalt chloride paper to change colour;		2
	(d)	• enzymes affected/to maintain enzyme activity/ref to body temperature being optimum for enzymes; • too high a temperature denatured/prevent denaturation; • too low, inactive/less active/prevent enzymes becoming inactive;		3

Total 13 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 5 of 14 · Question 3 part c.ii · 2 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part c.ii in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 6 of 14 · Question 6 part a · 3 marks

2024 JUN · Paper 02 · find part a in this mark scheme extract

6 (a)	• dim room or bright room/lamp at different distances; • allow time for (pupil) to adjust; • use a camera/video to take a picture/recording of the eye/ use of ruler; • measure the width/diameter of the pupil;	Ignore change light intensity	(3)
(b)	• diameter of pupil 6- 8 mm; • substitution 3.14 x radius of pupil; • accept any value between 28-51 mm^2;	If the diameter is used instead of the radius, then 2 marks can be awarded for any value between 113-201mm^2 Any value between 28-51 scores 3 marks	(3)
(c) (i)	• circular muscles relax; • radial muscles contract; • pupil gets larger/wider /dilates/ increase in size;		(3)
(ii)	• to allow more light to enter the eye; • to see image more clearly/better;		(2)

Total for Question 6 = 11 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 7 of 14 · Question 3 part b · 4 marks

2024 NOV · Paper 01 · extract 1 of 2 · find part b in this mark scheme extract

3 (a) (i)	Part W - pupil; Part X – retina;	1 mark for each correctly named part	2
(ii)	• sends (electrical) impulses from photoreceptors/rods/cones; • to the brain/CNS; OR • sends (electrical) impulses from brain/CNS; • to iris/ciliary muscles;		2
(iii)	lens becomes thinner/flatter/less convex; less refraction;	allow less light refracted	2
(iv)	Any two from: • shape of cornea restored; • light refracted, onto the retina/correctly/notmally; • correct information sent to the brain; • image/vision not blurred/clear;		2
(b)	• measure the size/diameter of the pupil; • in a dark room; • shine a light into the eye; • measure/compare size/diameter of the pupil;	accept ref to different light intensities	4

Total Marks 12

Type MS · Plan an investigation measuring a response or treatment in people

Answer 7 of 14 · Question 3 part b · 4 marks

2024 NOV · Paper 01 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 8 of 14 · Question 6 part b · 4 marks

2025 JUN · Paper 02 · extract 1 of 2 · find part b in this mark scheme extract

6	(a)	(i)	• A=rods/cones/retina; • detects light; • converts light energy into electrical energy; • (changed into (nerve) impulses; • B = transmits (nerve) impulses/electrical signals; • from (retina) to brain;	At least 1 mark from A and B to get full marks	4
		(ii)	• bent inwards at cornea/lens; • rays pass through the cornea and converge after the lens; • rays cross in eye (after lens); • inverted image/arrow;	Do not accept bending outwards for MP1 and MP2 Mp3- do not accept rays crossing before the retina	4
	(b)		• use both eyes and one eye; • use of eye patch/eq; • ball thrown same distance/stated distance;; • ball thrown many times; • count how many times ball is caught; • same type/size of ball;		4
	(c)		• misshapen/eq. cornea/lens; • light not focussed correctly/ light scatters/light not refracted properly; • blurred image;		2
					Total =14

Type MS · Plan an investigation measuring a response or treatment in people

Answer 8 of 14 · Question 6 part b · 4 marks

2025 JUN · Paper 02 · extract 2 of 2 · find part b in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 9 of 14 · Question 5 part a.i · 3 marks

2023 JAN · Paper 01 · extract 1 of 2 · find part a.i in this mark scheme extract

Number					
5	(a)	(i)	• use people from a range of ages; • hearing test using high frequency sounds; • control other variables e.g. sex, other health issues/same distance from sound source;		3
		(ii)	exposure to loud noises/genetic cause/infection		1
	(b)	(i)	• vibrations of the eardrum not transferred to the ossicles; • vibrations are not transferred to the cochlea; • hair cells are not stimulated/impulses not generated/not transferred to the auditory nerve/brain;		3
		(ii)	Two from: • conditions are moist/warm; • ideal conditions for bacterial growth/to multiply; • pathogens/bacteria more likely to get trapped;		2
		(iii)	Three from: • semi-circular canals; • contain fluid (in three tubes); • as the head moves so does the fluid; • transmits the information to the brain/cerebellum; • (cerebellum)co-ordinates a response to maintain balance/control balance;		3

Total 12 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 9 of 14 · Question 5 part a.i · 3 marks

2023 JAN · Paper 01 · extract 2 of 2 · find part a.i in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 10 of 14 · Question 5 part a · 4 marks

2022 JUN · Paper 01 · extract 1 of 2 · find part a in this mark scheme extract

5 (a)	• collect urine throughout the cold day; • measure the volume produced; • repeat for a hot day; • compare values;		4
(b)	any three from • vasodilation/arteriole widens; • increased flow of blood through skin; • heat lost through the skin's (surface); • by radiation;	allow artery	3
(c)	• (more) ADH released; • from pituitary gland; • makes kidney tubules/collecting ducts (more) permeable; • (more) water reabsorbed into the body/blood/from filtrate/urine;		4
			Total 11

Type MS · Plan an investigation measuring a response or treatment in people

Answer 10 of 14 · Question 5 part a · 4 marks

2022 JUN · Paper 01 · extract 2 of 2 · find part a in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 11 of 14 · Question 4 part d · 3 marks

2019 JUN · Paper 02 · extract 1 of 2 · find part d in this mark scheme extract

4 (a)	$\frac{30\,000}{100}$ x 40; 12 000;	Full marks for final correct answer ecf	1 1
(b)	a microorganism/named microorganism that causes disease;		1
(c)	It cannot be transmitted from one person to the next		1
(d)	• use a sample of people/use two groups of people; • feed each sample/group of people a different diet/feed one group a diet with vitamin B and the other without vitamin B; • monitor health/compare (health) of two groups;		1 1 1
(e)	• breakdown of cartilage/cartilage not formed properly/damaged; • bones rub together/more friction between bones/less shock absorption;		1 1

Total question 4 = 9 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 11 of 14 · Question 4 part d · 3 marks

2019 JUN · Paper 02 · extract 2 of 2 · find part d in this mark scheme extract

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 12 of 14 · Question 9 part a · 3 marks

2022 JUN · Paper 01 (regional R) · find part a in this mark scheme extract

9 (a)	• two groups of people; • provide one group with statins and the other without; • measure/compare levels of cholesterol / results after a period of time;		3
(b)	Any four from the following: • catheter inserted (into affected area); • balloon/stent inflated/expanded; • stent expands; • blood vessel opened/diameter of vessel increased; • blood flows more freely; • more oxygen to muscle (tissue); • more aerobic respiration;		4

Total marks 7

Question number	Answer	Notes	Marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 13 of 14 · Question 6 part b.iii · 5 marks

2024 JUN · Paper 02 (regional R) · find part b.iii in this mark scheme extract

6 (a)	• reduced blood flow to the heart/cardiac muscle/coronary arteries; • less oxygen to heart/cardiac muscle/tissue; • less (aerobic) respiration/less energy for contraction; • (cardiac) muscle/cells die; • (leading to) heart attack/heart disease;		(4)
(b) (i)	As the use of statins increases the level of blood cholesterol decreases;	allow inversely proportional	(1)
(ii)	1988 = 31; 2005 = 17; 1 : 1.8/1 : 0.54/1:0.548/1:0.55;	ecf for incorrect graph readings. if use 18:17 ratio is 1:1.0588 (accept 1:1.059, 1.06, 1.1) full marks for correct final answer	(3)
(iii)	• select sample of people; • record the number of people taking statins/not taking statins; • take blood samples; • measure the concentration/level of cholesterol (in blood); • repeat every year/between 1988-2008;		(5)
(iv)	any two from the following: • gender/age not taken into account; • sample size maybe too small; • reference to a named lifestyle not taken into account e.g. diet, level of activity/smoking; • genetic predisposition/diseases not taken into account;		(2)

Total for Question 6 = 15 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 14 of 14 · Question 5 part c · 6 marks

2022 JAN · Paper 02 · extract 1 of 2 · find part c in this mark scheme extract

number				
5 (a)		*near*	*distant*	
	lens	thick	thin;	
	circular ciliary muscle	contracted	relaxed;	
	suspensory ligaments	loose	stretched;	3
(b)	• two eyes; • each giving slightly different view; • allows better judgement of distances / depth;			3
(c)	any six from • test with two eyes then one; • use same person/people of similar age/same eye condition; • (throw and) catch a ball; • 10 times/suitable repeats; • record number of catches • ref control variable e.g. lighting/wind condition/same size ball/throw from the same distance;			6

Total 12 marks

Type MS · Plan an investigation measuring a response or treatment in people

Answer 14 of 14 · Question 5 part c · 6 marks

2022 JAN · Paper 02 · extract 2 of 2 · find part c in this mark scheme extract

Question number	Answer	Notes	Marks